

State Space Analysis Advantages

System Analysis Methods

State Space Analysis

- State Space method is applicable to Non Linear and Time Variant system.
- State Space method is applicable to MIMO.
- State Space method describes internal state of the system.
- State Space method gives zero state response.

Transfer function based Analysis

- Root locus
- Polar plot
- Nyquist plot
- Bode plot
- Routh stability

→ T.F. based analysis is only applicable to LTI system.

→ T.F. based Analysis is applicable to SISO.

→ T.F. based Analysis does not describe internal state of the system.

→ T.F. based Analysis does not give zero state response.

State Space Analysis Basics and Applications

→ It gives future behaviour of the system based on present Input and past history of the system.

[Future o/p is not based on prediction algo.]

→ The past history of the system described by STATE Variables.

[Here past history is initial states or Initial cond.ⁿ]

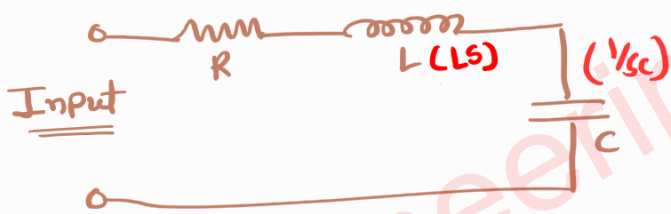
→ In Electrical circuit, no of State Variables depends on Inductors L and Capacitors C. It is not depending on resistor R.

→ No of State Variables can be defined by Order of the System.

$$\Rightarrow s^4 + 3s^3 + 2s^2 + 1 = 0$$

↑

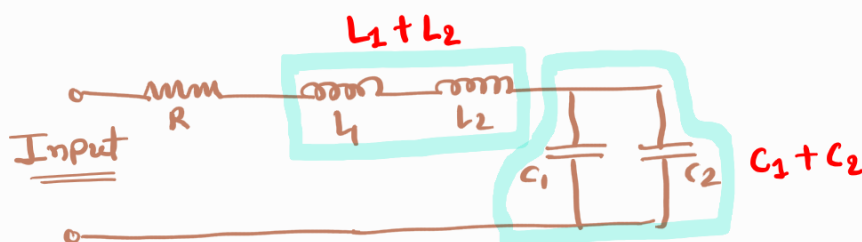
Order = 4, Hence Four state variables req.^{ed}



[Two State Variables]



[Four State Variables]



⇒ Applications

→ PLC

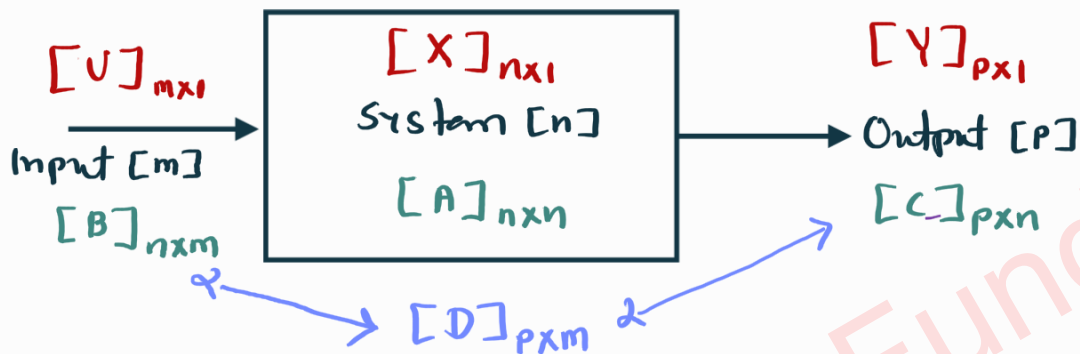
→ DCS

Representation, Terminology and Equations of State Space Analysis

→ Outlines

- Representation of State Space System
- Terms of State Space Analysis
- State eq.ⁿ & O/p eq.ⁿ of State Space System

→ State Space System



→ State eq.ⁿ / Dynamic eq.ⁿ

$$\Rightarrow \dot{X} = AX + BU$$

$$\Rightarrow [\dot{X}]_{nx1} = [A]_{nxn} [X]_{nx1} + [B]_{nxm} [U]_{mx1}$$

→ Output eq.ⁿ

$$\Rightarrow Y = CX + DU \quad \text{In most cases, } D \approx 0.$$

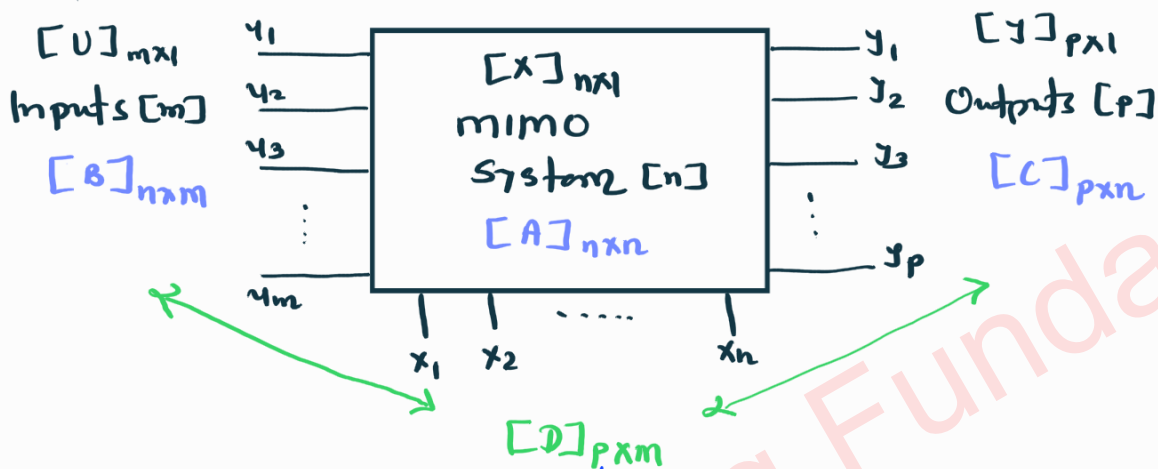
$$\Rightarrow [Y]_{px1} = [C]_{pxn} [X]_{nx1} + [D]_{pxm} [U]_{mx1}$$

State Space Analysis for MIMO System

Outlines

- mimo System
- State eq.ⁿ For mimo System
- Different models For State Space Analysis with mimo System

mimo System



State Space Eq.ⁿ

- State eq.ⁿ / Dynamic Eq.ⁿ → Output Eq.ⁿ
- $$\dot{X} = AX + BU$$
- $$Y = CX + DU$$

→ Different models For State Space Analysis with mimo System

- 1) Differential Eq.ⁿ Model
- 2) Transfer function Model
- 3) Signal Flow Graph
- 4) State Space For Electrical N/w

State Space Analysis to Identify Transfer Function

Determine the transfer function from the matrix of State Space Analysis

$$A = \begin{bmatrix} -3 & 1 \\ 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad C = [1 \ 1], \quad D = [0]$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
System Input Output Feedback

→ To Find Transfer function

$$G(s) = C[sI - A]^{-1}B + D$$

$$\begin{aligned} \rightarrow sI - A &= \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} -3 & 1 \\ 0 & -1 \end{bmatrix} \\ &= \begin{bmatrix} s+3 & -1 \\ 0 & s+1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \rightarrow [sI - A]^{-1} &= \frac{\text{Adj. } [sI - A]}{\text{Mag } [sI - A]} \\ &= \frac{\begin{bmatrix} s+1 & 1 \\ 0 & s+3 \end{bmatrix}}{s^2 + 4s + 3} \end{aligned}$$

$$\rightarrow G(s) = [1 \ 1] \frac{\begin{bmatrix} s+1 & 1 \\ 0 & s+3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}}{s^2 + 4s + 3} + 0$$

$$= [1 \ 1] \frac{\begin{bmatrix} s+2 \\ s+3 \end{bmatrix}}{s^2 + 4s + 3}$$

$$= \frac{s+2 + s+3}{s^2 + 4s + 3}$$

$$\rightarrow \boxed{G(s) = \frac{2s+5}{s^2 + 4s + 3}}$$

State Space Analysis to Find Transfer Function & Block Diagram

Determine Transfer Function for State Space eqⁿ

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 3 \\ -2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

→ Compare above eqⁿ with Standard eqⁿ

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

$$\rightarrow A = \begin{bmatrix} 0 & 3 \\ -2 & -5 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}, \quad D = 0$$

$\downarrow$ System $\downarrow$ Input $\downarrow$ Output $\downarrow$ Parameter

→ To get transfer function

$$K(s) = C[sI - A]^{-1}B + D$$

$$\rightarrow sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 3 \\ -2 & -5 \end{bmatrix}$$
$$= \begin{bmatrix} s & -3 \\ 2 & s+5 \end{bmatrix}$$

$$\rightarrow [sI - A]^{-1} = \frac{\text{Adj. } [sI - A]}{\text{Mag } [sI - A]}$$
$$= \frac{\begin{bmatrix} s+5 & 3 \\ -2 & s \end{bmatrix}}{s^2 + 5s + 6}$$

$$\rightarrow K(s) = \frac{\begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} s+5 & 3 \\ -2 & s \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 0}{s^2 + 5s + 6}$$
$$= \frac{\begin{bmatrix} 2s+8 & s+6 \\ s+5 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}}{s^2 + 5s + 6}$$
$$= \frac{\begin{bmatrix} 3s+14 & 3s+14 \\ s+8 & s+8 \end{bmatrix}}{s^2 + 5s + 6}$$

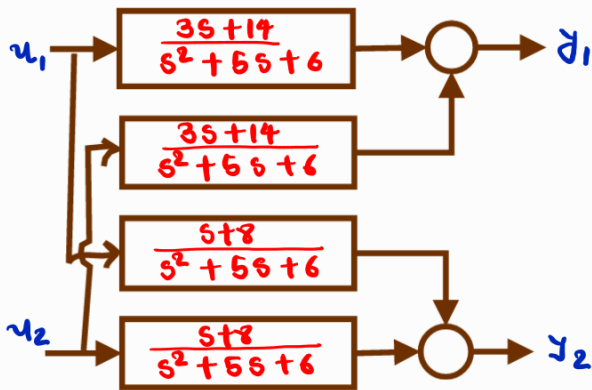
$$\rightarrow K(s) = \left[\frac{3s+14}{s^2+5s+6} \quad \frac{3s+14}{s^2+5s+6} \right]$$
$$\left[\frac{s+8}{s^2+5s+6} \quad \frac{s+8}{s^2+5s+6} \right]$$

$$\rightarrow Y(s) = K(s) U(s)$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \left[\frac{3s+14}{s^2+5s+6} \quad \frac{3s+14}{s^2+5s+6} \right] \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$
$$\left[\frac{s+8}{s^2+5s+6} \quad \frac{s+8}{s^2+5s+6} \right]$$

$$\rightarrow y_1 = \left[\frac{3s+14}{s^2+5s+6} \right] u_1 + \left[\frac{3s+14}{s^2+5s+6} \right] u_2$$

$$y_2 = \left[\frac{s+8}{s^2+5s+6} \right] u_1 + \left[\frac{s+8}{s^2+5s+6} \right] u_2$$



State Space Analysis to Identify Transfer Function

Find Transfer function of

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -5 & -1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix} r(t)$$

$$y = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

→ Standard eq.ⁿ for State Space

$$\dot{x} = Ax + Bu \quad \text{--- State eq.ⁿ}$$

$$y = Cx + Du \quad \text{--- Output eq.ⁿ}$$

$$\rightarrow A = \begin{bmatrix} -5 & -1 \\ 3 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 \end{bmatrix}, \quad D = 0$$

$\uparrow$ State $\uparrow$ Input $\uparrow$ Output $\uparrow$ Relavance

→ Transfer function is given by

$$G(s) = C[sI - A]^{-1}B + D$$

$$\rightarrow sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} -5 & -1 \\ 3 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} s+5 & 1 \\ -3 & s+1 \end{bmatrix}$$

$$\rightarrow [sI - A]^{-1} = \frac{\text{Adj. } [sI - A]}{\text{Mag } [sI - A]}$$

$$= \frac{\begin{bmatrix} s+1 & -1 \\ 3 & s+5 \end{bmatrix}}{s^2 + 6s + 8}$$

$$\rightarrow G(s) = \begin{bmatrix} 1 & 2 \end{bmatrix} \frac{\begin{bmatrix} s+1 & -1 \\ 3 & s+5 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix}}{s^2 + 6s + 8} + 0$$

$$= \frac{\begin{bmatrix} s+7 & 2s+9 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix}}{s^2 + 6s + 8}$$

$$= \frac{2s+14 + 10s+45}{s^2 + 6s + 8} = \frac{12s+59}{s^2 + 6s + 8}$$

Transfer Function to State Space Equation

A Feedback System is having the C.L.T.F.

$$T(s) = \frac{s^2 + 3s + 3}{s^3 + 2s^2 + 3s + 1}$$

Construct the State Model For the System.

$$\begin{aligned} \rightarrow T(s) &= \frac{Y(s)}{X(s)} = \frac{X(s)}{X(s)} \times \frac{Y(s)}{X(s)} \\ &= \left[\frac{1}{s^3 + 2s^2 + 3s + 1} \right] \left[s^2 + 3s + 3 \right] \end{aligned}$$

$$\rightarrow \frac{X(s)}{U(s)} = \frac{1}{s^3 + 2s^2 + 3s + 1} \quad \text{--- (A) [State eqn.]}$$

$$\rightarrow \frac{Y(s)}{X(s)} = s^2 + 3s + 3 \quad \text{--- (B) [o/p eqn.]}$$

→ From eqn. (A)

$$U(s) = s^3 X(s) + 2s^2 X(s) + 3s X(s) + X(s)$$

→ By taking Inverse Laplace

$$u(t) = \ddot{x} + 2\dot{x} + 3x + x$$

$$\left. \begin{aligned} x_1 &= x \\ x_2 &= \dot{x} = \dot{x}_1 \\ x_3 &= \ddot{x} = \dot{x}_2 \\ \dot{x}_3 &= \ddot{x} \end{aligned} \right\} \quad \text{--- (C)}$$

State eqn.
 $\dot{x} = Ax + Bu$

$$\Rightarrow u(t) = \dot{x}_3 + 2x_3 + 3x_2 + x_1$$

$$\Rightarrow \dot{x}_3 = -x_1 - 3x_2 - 2x_3 + u(t)$$

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ \dot{x}_3 &= -x_1 - 3x_2 - 2x_3 + u(t) \end{aligned}$$

$$\Rightarrow \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

→ From o/p eqn. (B)

$$\Rightarrow Y(s) = s^2 X(s) + 3s X(s) + 3X(s)$$

→ By taking Inverse Laplace

$$\Rightarrow y(t) = \ddot{x} + 3\dot{x} + 3x$$

$$\Rightarrow y(t) = x_3 + 3x_2 + 3x_1$$

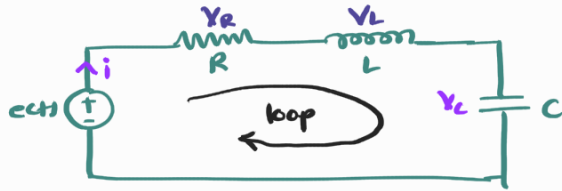
$$\Rightarrow y(t) = 3x_1 + 3x_2 + x_3$$

o/p eqn.
 $y = Cx + Du$

$$\rightarrow y(t) = \begin{bmatrix} 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} u(t)$$

State Space Analysis for Electrical System

Obtain State Space Model Considering Current and Capacitor Voltage as State Variable.



→ State eq.ⁿ
 $\dot{X} = AX + BU$

→ Output eq.ⁿ
 $Y = CX + DU$

$$\begin{aligned} - x_1 &= i \Rightarrow \dot{x}_1 = \frac{di}{dt} \\ x_2 &= V_C \Rightarrow \dot{x}_2 = \frac{dV_C}{dt} \end{aligned}$$

$$\rightarrow \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} R & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} c(t)$$

→ KVL in loop

$$\begin{aligned} \Rightarrow c(t) &= V_R + V_L + V_C \\ \Rightarrow c(t) &= iR + L \frac{di}{dt} + V_C \\ \Rightarrow c(t) &= x_1 R + L \dot{x}_1 + x_2 \\ \Rightarrow \dot{x}_1 &= -\frac{R}{L} x_1 - \frac{1}{L} x_2 + \frac{1}{L} c(t) \end{aligned}$$

→ Current through Capacitor

$$\begin{aligned} \Rightarrow i &= C \frac{dV_C}{dt} \\ \Rightarrow x_1 &= C \dot{x}_2 \\ \Rightarrow \dot{x}_2 &= \frac{1}{C} x_1 \end{aligned}$$

$$\rightarrow \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -R/L & -1/L \\ 1/C & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1/L \\ 0 \end{bmatrix} c(t)$$

→ If my outputs are V_R and V_C then

$$\begin{aligned} y_1 &= V_R = iR = R x_1 \\ y_2 &= V_C = x_2 \end{aligned}$$

State Space Modeling of Differential Equation

Make a State Space Model for given differential eq.ⁿ

$$F(t) = Kx(t) + b \frac{dx(t)}{dt} + m \frac{d^2x(t)}{dt^2}$$

Here o/p are distance $x(t)$ and Velocity $v(t)$.

$$\begin{aligned} \rightarrow x_1 &= x(t) \\ x_2 &= \frac{dx(t)}{dt} = \dot{x}_1 \\ \dot{x}_2 &= \frac{d^2x(t)}{dt^2} \end{aligned}$$

→ State eq.ⁿ
 $\dot{X} = AX + BU$
 → Output eq.ⁿ
 $Y = CX + DU$

→ So, Force eq.ⁿ

$$\begin{aligned} \Rightarrow F(t) &= Kx_1 + b\dot{x}_2 + m\ddot{x}_2 \\ \Rightarrow \dot{x}_2 &= -\frac{K}{m}x_1 - \frac{b}{m}\dot{x}_2 + \frac{1}{m}F(t) \end{aligned} \quad \left\{ \begin{aligned} \Rightarrow \dot{x}_1 &= x_2 \end{aligned} \right.$$

→ State eq.ⁿ

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -K/m & -b/m \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1/m \end{bmatrix} F(t)$$

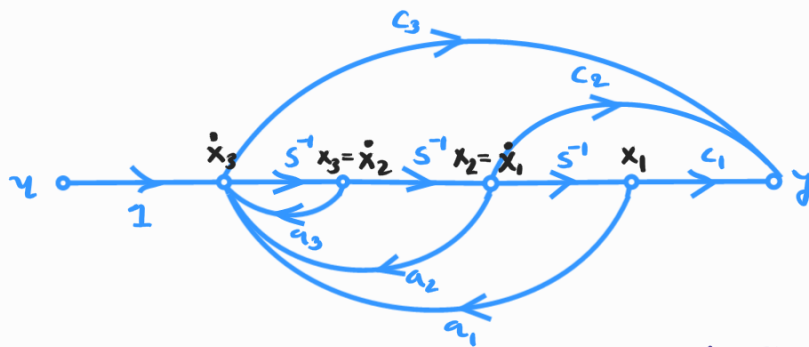
→ o/p

$$\begin{aligned} y_1 &= x(t) = x_1 \\ y_2 &= v(t) = \frac{dx(t)}{dt} = x_2 \end{aligned}$$

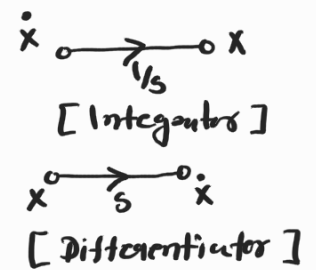
→ O/p eq.ⁿ

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} F(t)$$

State Space Analysis for Signal Flow Graph



→ State eqⁿ
 $\dot{x} = Ax + Bu$
 → Output eqⁿ
 $y = Cx + Du$



Derive State Model of given Signal Flow graph.

State eqⁿ

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = a_1x_1 + a_2x_2 + a_3x_3 + u$$

→

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} [u]$$

$\underline{\dot{x}} \quad \quad \underline{A} \quad \quad \underline{x} \quad + \quad \underline{B} \quad \quad \underline{u}$

Output Equation

$$y = c_1x_1 + c_2x_2 + c_3x_3$$

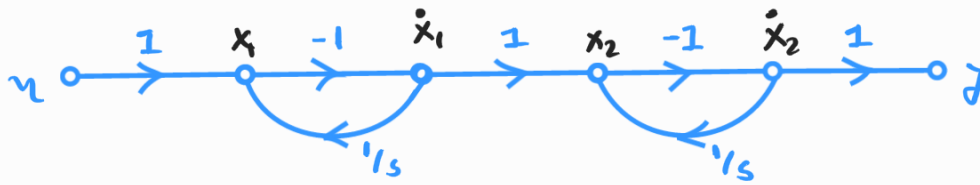
$$y = c_1x_1 + c_2x_2 + c_3a_1x_1 + c_3a_2x_2 + c_3a_3x_3 + c_3u$$

$$y = (c_1 + c_3a_1)x_1 + (c_2 + c_3a_2)x_2 + c_3a_3x_3 + c_3u$$

$$\begin{bmatrix} y \end{bmatrix} = \begin{bmatrix} c_1 + c_3a_1 & c_2 + c_3a_2 & c_3a_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} c_3 \end{bmatrix} [u]$$

$\underline{y} \quad \quad \underline{C} \quad \quad \underline{x} \quad + \quad \underline{D} \quad \quad \underline{u}$

State Space Analysis for Signal Flow Graph



- State eq.ⁿ

$$\dot{x} = Ax + Bu$$

- Output eq.ⁿ

$$y = Cx + Du$$

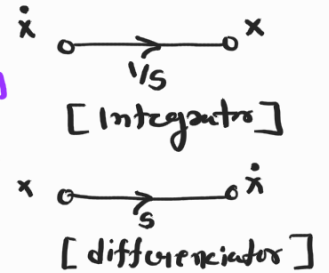
- state transition matrix

$$\phi = e^{At} = \mathcal{L}^{-1}[(sI - A)^{-1}]$$

Derive State Model of given Signal Flow graph.

State eq.ⁿ

$$\begin{aligned} \rightarrow \dot{x}_1 &= -x_1 = -[u + \frac{1}{s}\dot{x}_1] \\ &= -[u + x_1] \\ &= -x_1 - u \\ \rightarrow \dot{x}_2 &= -x_2 \\ &= -[\dot{x}_1 + \frac{1}{s}\dot{x}_2] \\ &= -[-x_1 - u + x_2] \\ &= x_1 - x_2 + u \end{aligned} \quad \left| \begin{aligned} \rightarrow \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u \\ &\quad \underline{\dot{x}} \quad \quad \underline{A} \quad \quad \underline{x} \quad + \quad \underline{B} \quad \underline{u} \\ \rightarrow \text{Output eq.ⁿ} \\ \rightarrow y &= \dot{x}_2 \\ &= x_1 - x_2 + u \\ \rightarrow \underline{y} &= \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \end{bmatrix} u \\ &\quad \underline{C} \quad \quad \underline{x} \quad + \quad \underline{D} \quad \underline{u} \end{aligned} \right.$$



State Transition matrix

$$\phi = \mathcal{L}^{-1}[(sI - A)^{-1}]$$

$$\begin{aligned} - sI - A &= \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} s+1 & 0 \\ -1 & s+1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} - [sI - A]^{-1} &= \frac{\text{Adj. } [sI - A]}{\det. [sI - A]} \\ &= \frac{\begin{bmatrix} s+1 & 0 \\ 1 & s+1 \end{bmatrix}}{(s+1)^2} \\ &= \begin{bmatrix} \frac{1}{s+1} & 0 \\ \frac{1}{(s+1)^2} & \frac{1}{s+1} \end{bmatrix} \end{aligned}$$

$$- \phi = \mathcal{L}^{-1}[(sI - A)^{-1}]$$

$$= \begin{bmatrix} e^{-t} & 0 \\ te^{-t} & e^{-t} \end{bmatrix}$$

Controllability & Observability using Kalman's Test

Basics of Controllability and Observability

- A system is controllable if it's possible to drive the system from any initial state to any desired final state in a finite amount of time using suitable control inputs.
- If the system's state can be determined from the knowledge of the system's Output, then the system is observable.

Question 1 – Check Controllability and Observability and also compute the number of controllable and observable states for the following system.

$$\Rightarrow \dot{X} = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix} X + \begin{bmatrix} 1 \\ -1 \end{bmatrix} U \quad \sim \text{State eq.}^n \Rightarrow \dot{X} = AX + BU$$

$$\Rightarrow Y = [1 \quad 1]X \quad \sim \text{O/p eq.}^n \Rightarrow Y = CX + DU$$

$$\sim A = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad C = [1 \quad 1], \quad D = 0$$

→ Controllability Matrix Q_c ($m \times m$) (2×2)

$$\begin{aligned} \rightarrow Q_c &= [B \quad AB \quad A^2B \quad A^3B \quad \dots \quad A^{m-1}B] \\ &= [B \quad AB] \\ &= \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \end{aligned}$$

→ $\text{mag } Q_c = 0 \Rightarrow \text{Rank} \neq \text{Order}$. (given system is not controllable).

→ identity sub matrix Q_c $(m-1) \times (m-1)$ order 1×1

$$\begin{aligned} -Q_{c11} &= [1], \quad Q_{c12} = [-1], \quad Q_{c21} = [-1], \quad Q_{c22} = [1] \\ &\neq 0 \end{aligned}$$

→ Rank of $Q_c = m-1 = 1$

→ Controllable states = Rank of $Q_c = 1$ & 1 state uncontrollable.

→ Observability Matrix Q_o [$m \times m$]

$$\begin{aligned} \rightarrow Q_o &= \begin{bmatrix} C \\ AC \\ A^2C \\ \vdots \\ A^{m-1}C \end{bmatrix} = \begin{bmatrix} C \\ AC \end{bmatrix}_{2 \times 2} = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \end{aligned}$$

$$\rightarrow A = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}, \quad C = [1 \quad 1]$$

→ $\text{mag } Q_o = 0$, Rank of $Q_o \neq \text{Order}$.

→ Sub Matrix Q_o with Order $(m-1) \times (m-1) = (1 \times 1)$

$$\begin{aligned} Q_{o11} &= [1], \quad Q_{o12} = [1], \quad Q_{o21} = [-1], \quad Q_{o22} = [-1] \\ &\neq 0 \end{aligned}$$

→ Rank of $Q_o = m-1 = 1$

→ Observable state = Rank of $Q_o = 1$ & 1 unObservable state.

Question 2 – Check Controllability and Observability and also compute the number of controllable and observable states for the following system.

$$\Rightarrow A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 4 \\ 0 \end{bmatrix}, C = [1 \quad 1 \quad 1]$$

~> Controllability Matrix Q_c (3×3)

$$Q_c = [B \quad AB \quad A^2B] \quad - \quad A^2B = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 4 \\ 0 \end{bmatrix}$$
$$= \begin{bmatrix} 0 & 4 & -8 \\ 4 & -4 & 4 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$$

~> $\text{Mag } Q_c = 0 \Rightarrow \text{Rank of } Q_c \neq \text{Order of } Q_c$, System is not Controllable.

~> Sub matrix Q_c ($(m-1) \times (m-1)$)
 2×2

$$Q_c' = \begin{vmatrix} 4 & -8 \\ -4 & 4 \end{vmatrix} = 16 - 32 = -16 \neq 0$$

$$\leadsto \text{Rank of } Q_c = m-1 = 3-1 = 2$$

$$\leadsto \text{Controllable states of } Q_c = \text{Rank of } Q_c \\ = 2 \text{ \& 1 state is uncontrollable.}$$

$\leadsto$ Observability Matrix Q_o ($m \times m$) (3×3)

$$\Rightarrow Q_o = \begin{bmatrix} C \\ AC \\ A^2C \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \quad C = [1 \ 1 \ 1]$$

$$\Rightarrow Q_o = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & -2 \\ 1 & -1 & 4 \end{bmatrix}$$

$$-A^2C = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix} [1 \ 1 \ 1]$$

$$\leadsto \det Q_o = 1(-2) - 1(-4+2) + 1(1) \\ = 1 \neq 0$$

$\leadsto$ Hence, System is observable, 3 states are observable.